SEEING THROUGH THE VEIL; SATELLITE SUPER-RESOLUTION FOR PLANT PHENOTYPING AND THE FUTURE OF REMOTE SENSING

Priority Area(s): Data-fusion techniques for bio-energy crops, super-resolution for trait quantification, sub-pixel shifts for spatial and spectral interpolation.

Overview

Second generation bio-energy feedstock crops such as miscanthus, sorghum, and sugar cane are expensive and labor intensive to phenotype at scale. Over the past fifteen years, remote sensing via uncrewed aerial vehicles (UAVs) has emerged as a dominant method for standardized, scalable, nondestructive collection of phenotypic traits. However, remote sensing is still susceptible to human error, reliant of heavy startup costs and cooperative weather patterns. To overcome this, research has begun on enhancing satellite imagery to achieve comparable synthetic spatial resolutions from satellite imagery.

Proposal

I propose a multi-image super-resolution (MISR) approach to achieving sub-pixel trait quantification, using small frame-shifts in alignment to calculate individual traits. Additionally, I would like to fuse this with a comparison of spectral unmixing and diffusion methods for edge refinement. The purpose of this is to understand whether perceptually accurate methods such as spectral unmixing or diffusion can polish rough interpolations in a way that specifically targets improved downstream yield predictions.

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